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RESEARCH INTERESTS

Computing Education, Computing Education Curriculum, Teaching and Learning Methods, Evaluation Methods, Bebras Challenge, Informatics Gifted, Teacher Training

EDUCATION

PhD in Computer Education, Feb. 2017 – Korea National University of Education.
Dissertation: The Development and Application of a CT-CPS(Computational Thinking-based Creative Problem Solving) Instructional Model for the Software Education of New Curriculum.

MA in Computer Education, Feb. 2014 – Korea National University of Education.

BA in Computer Education, Feb. 2005 – Korea National University of Education.

APPOINTMENTS

Associate Professor: Andong National University, Department of Computer Education, Sep. 2022 to Present.

Assistant Professor: Andong National University, Department of Computer Education, Sep. 2018 to Aug. 2022.

Visiting Scholar: University of Florida, Aug. to Jul. 2025

Teacher: Ministry of Education(Chungbuk), Mar. 2005 to Aug. 2018

PUBLICATIONS

Book

- Yongju Jeon, Hyeonjong Choi. *Informatics Subject Pedagogy, 3rd edition*. Seoul: Hanbit Academy, 2026
- Yongju Jeon, et al. *School Education System in the Future Intelligent Information Society: Analysis and Prospects*. Education Science Press, 2025
- Yongju Jeon. *Web Design & Programming based on Responsive Web: Bootstrap 5*. Gyeongbuk: Dongin Press, 2022
- Yongju Jeon, et al. *Artificial Intelligence Basic(Textbook of High school)*. Seoul: Seongandang Press, 2021
- Yongju Jeon, et al. *Artificial Intelligence and Physical Computing(Textbook of High school)*. Seoul:Geumseong Press, 2020
- Yongju Jeon, et al. *Learning Computer Science with Bebras Challenge*. Seoul: Saengneung Press, 2019

Peer-reviewed Journals

- Yongju Jeon, Shinok Kang, Dasom Park, Jaehyoun Kim. "Analysis of Requirements for AI Education by Stakeholders and Exploration of Directions for Future Curriculum Improvement", *THE JOURNAL OF KOREAN ASSOCIATION OF COMPUTER EDUCATION*, 29(8), 134-149, 2026.
- Heejeong Park, Yongju Jeon, Chanjoo Lee. "Effectiveness Analysis of a Digital Competency Enhancement Professional Development Program for Primary and Secondary School Teachers: Focusing on Teachers in Gyeongbuk", *Journal of Creative Information Culture*, 12(2), 215-228, 2026
- Meize Guo, Kristen Apraiz, Yongju Jeon, Michael J. Johnson, Gayle Evans, and Maya Israel. "Robot Sharks: An AI and STEM Adventure for 5th Grade Student", *Journal of Technology-Integrated Lessons and Teaching*, 4(2), 50-61, 2025
- Heejeong Park, Yongju Jeon, Chanjoo Lee. "Competency of Primary and Secondary School Teachers: Focusing on Teachers in Gyeongbuk", *Journal of Education & Culture*, 30(6), 71-104, 2024
- Heejeong Park, Yongju Jeon. "The Development and Application of a MCPS (Motivation and Creative Problem Solving) Instructional Model in Computing Liberal Arts for Non-Majors", *International Journal on Advanced Science, Engineering and Information Technology*, 14(5), 1534-1552, 2024

- Seulki Kim, Yongju Jeon. "Development of a Python Library to Generate Synthetic Datasets for Artificial Intelligence Education", *International Journal on Advanced Science, Engineering and Information Technology*, 14(3), 936-945, 2024
- Juyoung Song & Yongju Jeon. "Development and Application of Artificial Intelligence Ethics Education Program for Elementary School Students based on Hans Jonas' Responsibility Ethics", *Journal of Korean Association of Computer Education*, 26(2): 29-39, 2023
- Seulki Kim, Yongju Jeon, Taeyoung Kim. "Research on the Development and Utility Analysis of K-12 Artificial Intelligence Educational Datasets Using Synthetic Datasets Generation Method", *Journal of Korean Association of Computer Education*, 25(3): 9-21, 2022
- Heejeong Park & Yongju Jeon. "A Design and Application of Software Liberal Arts Course based on CT-CPS Model for Developing Creative Problem-Solving Ability and Learning Motivation of Non-software Majors", *International Journal of Informatics Visualization*, 6(2): 317-326, 2022
- Ohhan Kang, Heejeong Park, Yongju Jeon. "Analysis of the Informatics & Computer Pedagogical Questions in the Test for the Appointment of Secondary Teachers", *Journal of Korean Association of Computer Education*, 25(2): 13-21, 2022
- Yongju Jeon. "Analysis of Learning Objectives and Contents of High School 'Basic Artificial Intelligence' Textbook", *Journal of Korean Association of Computer Education*, 25(1): 1-16, 2022
- Yongju Jeon, Jamee Kim, Hanin Kim. "Subjects Composition and Content Knowledge System of Informatics for the 2022 Revised National Curriculum", *Journal of Korean Association of Computer Education*, 24(6): 1-16, 2021
- Sujin Jun, Yongju Jeon, Inkee Jeong. "Design of Liberal Arts Subjects for Artificial Intelligence Education for Pre-Teachers in Elementary and Secondary Schools", *Journal of the Korean Association of Information Education*, 25(5): 859-869, 2021
- Heejeong Park, Hyoseon Kim, Jeongim Choi, Yongju Jeon. "Development of Teaching Efficacy Instrument in Informatics(Software and AI) Subject", *Journal of Korean Association of Computer Education*, 24(4): 39-52, 2021
- Seulki Kim, Yongju Jeon, Taeyoung Kim. "A Study on Data Preprocessing Content Knowledges According to School Level for Artificial Intelligence Education", *Journal of Korean Association of Computer Education*, 24(4): 1-12, 2021

Heeok Heo, Shinccheon Kang, Yongju Jeon, Hanil Kim, "The Suggestion of Big Ideas for the Next Informatics Curriculum", *Journal of Korean Association of Computer Education*, 24(2): 39-51, 2021

Seyoung Park, Yongju Jeon, Jeonghee Seo, "Development and Effectiveness Analysis of Training Program for Core Teachers of Elementary SW Education", *Journal of Korean Association of Computer Education*, 23(3): 31-40, 2020

Yongju Jeon, "Sequence Analysis and Suggestion for Linking Contents of Elementary and Secondary Software Education", *Journal of Creative Information Culture*, 5(2): 105-116, 2019

Yongju Jeon, "Exploration of the Direction of Customizing HUSTOJ-based Open Source Online Judge Systems for Informatics Education", *Journal of Creative Information Culture*, 5(1): 53-61, 2019

Yongju Jeon, Ungyeol Jung, Injoo Kim, Jihyea Kim, Hyuna Lee, Dongyoon Kim, "An Exploratory Study on Teaching & Learning and Evaluation Methods using Beaver Challenge in Software Education", *Journal of Korean Association of Computer Education*, 21(6): 63-82, 2018

Yongju Jeon, Kisang Song, "Short-term ICT Training Program for Non-Computer Science Major Teachers in Developing Countries for Improving ICT Teaching Efficacy", *The International Journal of Advanced Smart Convergence*, 7(2): 73-85, 2018

Yongju Jeon, Taeyoung Kim, "The Development and Application of a Responsive Web-based Smart Learning System for the Cyber Project Learning of Elementary Informatics Gifted Students", *Journal of Theoretical and Applied Information Technology*, 96(5): 1387-1397, 2018

Proceedings

Meize Guo, Yongju Jeon, Alexis Cobo, Kristen Apraiz, Maya Israel. "Exploring Preservice Teachers' Conceptual Trajectories While Teaching an AI and STEM Integrated Unit: A Research-in-Practice Report", WiPSCE '26: Proceedings of the 20th WiPSCE Conference on Primary and Secondary Computing Education Research, 16, 1-6. 2026

Jun Hee Lee, Yongju Jeon, "Methods for Applying Lessons for the Initial Understanding of Artificial Intelligence of Students with Developmental Disabilities", *Proceeding of Korean Association of Computer Education Conference*, 28(1), 125-128, 2024.

Yongju Jeon et al., "Analysis of the Connections between Bebras Challenge 2022 Tasks and the 2015 and 2022 Revised Informatics National Curriculum", *Proceeding of Korean Association of Computer Education Conference*, 27(1), 39-42, 2023.

Jaeman Jo, Yongju Jeon, "A Data Literacy Education Program Using Scientific Data", *Proceeding of Korean Association of Computer Education Conference*, 27(1), 315-317, 2023.

Heejung Park, Yongju Jeon, "Fundamental Study for the Development of University Software Liberal Arts Class Model(MBA_PD) for Non-majors", *Proceeding of Korean Association of Computer Education Conference*, 26(2), 51-54, 2022.

Seulki Kim, Yongju Jeon, Taeyoung Kim, "A Study on the Development of Data Sets for Artificial Education Using Synthetic Datasets Generation Method", *Proceeding of Korean Association of Computer Education Conference*, 26(1), 227-230, 2022.

Yongju Jeon, Dongyoon Kim, "Analysis of the Connection between Bebras Challenge 2021 Tasks and 2015 Revised Informatics National Curriculum", *Proceeding of Korean Association of Computer Education Conference*, 26(1), 85-88, 2022.

Heejung Park, Yongju Jeon, "A Design of SW Liberal Arts Course based on CT-CPS Model for Developing Creative Problem-Solving Ability and Learning Motivation of Non-SW majors", *International Conference on Interdisciplinary Research on Computer Science, Psychology, and Education(ICICPE)*, 2021.

Yongju Jeon et al., "A Proposal of Data Set Analysis and Application for AI Education in Primary and Secondary school", *Proceeding of Korean Association of Computer Education Conference*, 25(1), 55-58, 2021.

Hyunbeom Choi, Yongju Jeon, Wonseog Kang, "Development and Application of Hackathon Making Program for AI Education in Elementary School Students", *Proceeding of Korean Association of Computer Education Conference*, 25(1), 143-145, 2021.

Hyoseon Kim, Heejung Park, Jeongim Choi, Yongju Jeon, "Fundamental Study for the Development of Teaching Efficacy Instrument in Informatics(Software and AI) Subject",

Proceeding of Korean Association of Computer Education Conference, 25(1), 105-108, 2021.

Yongju Jeon, UngYeol Jung, "Development of the Educational Case through Alignment of Curriculum-Instruction-Evaluation based on Bebras challenges", *Proceeding of Korean Association of Computer Education Conference, 24(2), 35-38, 2020.*

Yongju Jeon, UngYeol Jung, "Analysis of the Tasks of Bebras ChallengeApplicable to Big Data Analytics Education", *Proceeding of Korean Association of Computer Education Conference, 24(2), 31-34, 2020.*

Yongju Jeon, Sujin Jun, UngYeol Jung, "Analysis of the Connection between Bebras Challenge 2019 Tasks and Achievement Standards of 2015 Revised Software National Curriculum", *Proceeding of Korean Association of Computer Education Conference, 24(1), 73-76, 2020.*

Yongju Jeon, UngYeol Jung, "An Exploratory Study on Tasks of Bebras Challenge for Artificial Intelligence Education", *Proceeding of Korean Association of Computer Education Conference, 24(1), 69-72, 2020.*

Yongju Jeon, "Suggestions for the Direction of Application for Informatics Assessment through the Research of Project Quantum Questions", *Proceeding of Korean Association of Computer Education Conference, 23(2), 69-71, 2019.*

Yongju Jeon, Jeong Hee Seo, "Development of Training Program for Elementary SW Education Leader Teachers", *Proceeding of Korean Association of Computer Education Conference, 23(2), 59-62, 2019.*

Yongju Jeon et al., "How to connect between abstraction and automation for CT education - focused on Korean SW education", *Proceedings of CSTA 2019 Annual Conference, 2019.*

Injoo Kim, Seulki Kim, Yongju Jeon, "A Study on Application of Formative Assessment using Bebras Challenge in Software Education", *Proceeding of Korean Association of Computer Education Conference, 23(1), 55-58, 2019.*

Yongju Jeon, "A Study on Utilization of Bebras Challenge Tasks as a Measurement Tool for Computational Thinking", *Proceeding of Korean Association of Computer Education Conference, 23(1), 48-52, 2019.*

Yongju Jeon, Taeyoung Kim, "The Design and Implementation of Responsive Web-based Smart Learning System for Elementary Informatics Gifted Students' Cyber Project Learning", ICCCA2017 Proceedings, 2017.

Yongju Jeon et al., "A Case Study on the Application of the Computational Thinking-based Creative Problem Solving (CT-CPS) Instructional Model", Edulearn16 Proceedings, 2016.

Yongju Jeon, Taeyoung Kim., "jQuery Mobile As A Text-based Introductory Educational Programming Language", Edulearn15 Proceedings, 2015.

RESEARCH PROJECT EXPERIENCE

Improving the System for Retraining In-service Teachers to Strengthen AI and Digital Competencies, 4th Branch of AI Education Alliance and Policy(AIEDAP), Ministry of Education, Sep. 2022 to 2024. (Co Researcher)

Training of SW-AI Education Competency Building for In-service Teachers (Artificial Intelligence Basics), Gyeongbuk Office of Education, Jul. 2022 to Nov. 2022. (Project Manager)

Reimagining School Knowledges and Restructuring Curriculum based on Megatrends of the Future Society, Korea Institute for Curriculum and Evaluation(KICE), Mar. 2022 to Dec. 2022. (Co Researcher)

2022 Revised National Informatics Curriculum Development Research. Ministry of Education. Nov. 2021 to Dec. 2022. (Co Researcher)

How to Improve Equity of Opportunity in Elementary and Secondary SW and AI Education. Korea Foundation for the Advancement of Science & Creativity(KOFAC). Oct. 2021 to Dec. 2021. (Co Researcher)

Training of Gifted Education Training for K-12 Teachers of GyeongBuk(Software Basic), Gyeongbuk Office of Education, Nov. 2020 to Feb. 2021. (Project Manager)

Analyzing and utilizing datasets for K-12 AI education. Korea Education and Research Information Service(KERIS). May 2020 to Nov. 2020. (Project Manager)

Training of Gifted Education Training for K-12 Teachers of GyeongBuk(Software Basic), Gyeongbuk Office of Education, Nov. 2019 to Feb. 2020. (Project Manager)

TEACHING EXPERIENCE

Assistant/Associate Professor, Andong National University

- Informatics Subject Pedagogy
- Informatics Subject Instructional Methods
- Informatics Subject Logics and Essays
- Informatics Subject Material Research and Teaching Practice
- Informatics Subject Evaluation Methods
- Linux Programming
- Computer Programming
- AI and Data Analysis Basics

Graduate Teaching Assistant, Korea National University of Education

- ICT Education
- C Programming
- Android Programming
- Physical Computing Basics
- Understanding and Development of Mobile Homepages

HONORS / AWARDS

Award of Merit for Outstanding Achievement in Promoting National Curriculum Policy by the Minister(Ministry of Education), 2023

Award of Best Paper of the Year in Computer Education(Journal of Korean Association of Computer Education), 2021 and 2022

GKNU Excellence in Education Award, 2021

Award of Merit for Software Education Development by the Minister(Ministry of Science and ICT), 2017

MEMBERSHIPS / AFFILIATIONS

Korean Association of Computer Education

Korean Association of Information Education

International Bebras Organization

Bebras Korea

Head of Future Education Center (GKNU)

Head of Software Convergence Education (GKNU)